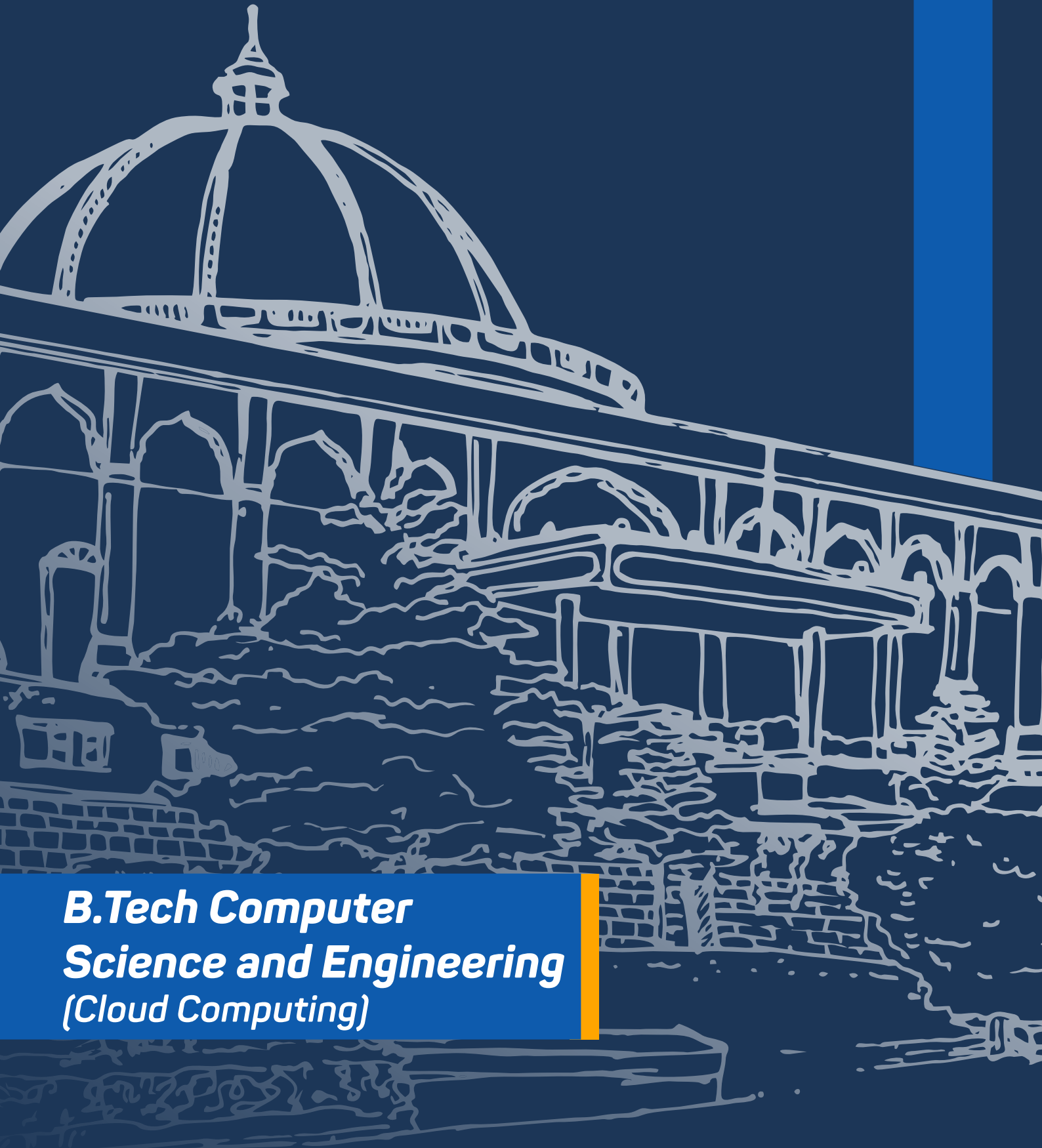




***B.Tech Computer
Science and Engineering
(Cloud Computing)***

Division	Faculty of Engineering and Technology
School Name	School of Computer Science & Engineering
Department Name	Department of Computer Engineering and Technology
Programme Name	B.Tech Computer Science and Engineering (Cloud Computing)

Semester	Odd (I)	Even (II)	Total Credits
First Year	20	21	41
Second Year	22	22	44
Third Year	22	22	44
Fourth Year	22	14	36

+ + + + + + + + + + + + + + **COURSE BASKET** + + + + + + + + + + + + + +

| Course Type | Description |
|----------------------|--|
| Programme Core | Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU |
| Programme Electives | Open electives under the programme allow students to specialise in a particular area connected to their major. |
| University Core | Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students. |
| University Electives | Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core. |

Semester I

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| I | UC | Indian Constitution | 1 |
| I | UC | Environment and Sustainability | 1 |
| I | UC | Yoga - I | 1 |
| I | UC | Social Leadership Development Program | 1 |
| I | UC | Financial Literacy | 1 |
| I | UC | Python Programming | 2 |
| I | PF | Linear Algebra and Differential Calculus | 3 |
| I | PF | Engineering Chemistry | 3 |
| I | PF | Engineering Physics | 3 |
| I | PF | Engineering Graphics | 3 |
| I | PF | Ideas and Innovations in Manufacturing | 1 |
| | | Total Credits: | 20 |

Semester II

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| II | UC | Yoga - II | 1 |
| II | UC | Co-creation | 1 |
| II | UC | Foundation of Peace | 2 |
| II | UC | Indian Knowledge System (General) | 2 |
| II | UC | Sports | 1 |
| II | PF | Engineering Mechanics | 3 |
| II | PF | Foundations of Programming | 4 |
| II | PF | Discrete Mathematics with Graph Theory | 3 |
| II | Program Capstone Project/
Seminar and Internships | Foundations of Computer Architecture
and System Design | 4 |
| | | Total Credits: | 21 |

Semester - III

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| III | UC | Spiritual and Cultural heritage: Indian Experience | 2 |
| III | UC | Research Innovation Design Entrepreneurship (RIDE) | 1 |
| III | UE | University Electives - I | 3 |
| III | PF | Calculus and Numerical Methods | 4 |
| III | PF | Data Structures | 3 |
| III | PF | Data Structures Laboratory | 1 |
| III | PM | Operating System | 3 |
| III | PM | Operating System Laboratory | 1 |
| III | PM | Computer Networks | 3 |
| III | PM | Computer Networks Laboratory | 1 |
| | | Total Credits: | 22 |

SEMESTER - IV

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| IV | UC | Rural Immersion | 1 |
| IV | UC | Life Transformation Skills | 1 |
| IV | UE | University Electives - II | 3 |
| IV | Program Capstone Project/
Seminar and Internships | Object Oriented Concepts using C++ | 2 |
| IV | PF | Probability and Statistics | 4 |
| IV | PM | Database Management System | 3 |
| IV | PM | Database Management System Laboratory | 1 |
| IV | PF | Virtualization & Cloud Computing | 3 |
| IV | Program Capstone Project/
Seminar and Internships | Cognitive Skills | 1 |
| IV | PF | Indian Knowledge System (Sci. & Tech.) | 2 |
| IV | Program Capstone Project/
Seminar and Internships | Full Stack Development Laboratory | 1 |
| | | Total Credits: | 22 |

Semester - V

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| V | UC | Managing Conflicts Peacefully: Tools and Techniques | 2 |
| V | UE | University Electives - III | 3 |
| V | PE | Program Elective - I | 4 |
| V | PM | Cloud Computing Architecture | 4 |
| V | PM | Design and Analysis of Algorithms | 3 |
| V | Program Capstone Project/Seminar and Internships | Design and Analysis of Algorithms Laboratory | 1 |
| V | PM | Cloud Platforms | 2 |
| V | PM | Theory of Computation | 3 |
| | | Total Credits | 22 |

Semester - VI

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| VI | UC | National Academic Immersion Program | 2 |
| VI | PE | Program Elective - II | 4 |
| VI | PM | Cloud Privacy & Security | 4 |
| VI | PM | Data Engineering and Data Visualization | 4 |
| VI | PM | Containerization & Kubernetes | 4 |
| VI | PF | Cloud-Native Application Design | 1 |
| VI | Program Capstone Project/Seminar and Internships | Competitive Coding | 1 |
| VI | Program Capstone Project/Seminar and Internships | Mini Project using Java Programming | 1 |
| VI | Program Capstone Project/Seminar and Internships | Seminar | 1 |
| | | Total Credits: | 22 |

Semester - VII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| VII | PE | Program Elective - III | 4 |
| VII | PM | Big Data Processing in Cloud | 3 |
| VII | PM | Edge Computing & Fog Computing | 3 |
| VII | PM | DevSecOps | 3 |
| VII | Program Capstone Project/Seminar and Internships | Capstone Project | 6 |
| VII | PM | Artificial Intelligence and Machine Learning Techniques | 3 |
| | | Total Credits: | 22 |

Semester - VIII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|----------------------------|---------------|
| VIII | PE | Program Elective - IV | 4 |
| VIII | Program Capstone Project/Seminar and Internships | Internship | 10 |
| | | Total Credits: | 14 |

Program Electives

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|------------------------|---|---------------|
| V | Program Elective - I | A. Bigdata Technologies | 4 |
| V | Program Elective - I | B. Computer Graphics and 3D Modelling | 4 |
| V | Program Elective - I | C. Wireless and Mobile Networks | 4 |
| V | Program Elective - I | D. AR & VR | 4 |
| VI | Program Elective - II | A. IoT Application in Cloud | 4 |
| VI | Program Elective - II | B. Digital Enterprise Management | 4 |
| VI | Program Elective - II | C. GCP (Google Cloud Platform) | 4 |
| VI | Program Elective - II | D. Natural Language Processing (NLP) | 4 |
| VII | Program Elective - III | A. Data Visualization and Business Intelligence (BI) | 4 |
| VII | Program Elective - III | B. Disaster Recovery and Business Continuity in Cloud | 4 |
| VII | Program Elective - III | C. Quantum Computing in Cloud | 4 |
| VII | Program Elective - III | D. High-Performance Computing (HPC) in Cloud | 4 |
| VIII | Program Elective - IV | A. Blockchain | 4 |
| VIII | Program Elective - IV | B. Cloud Governance, Risk, and Compliance (GRC) | 4 |
| VIII | Program Elective - IV | C. Deep Learning and Generative AI | 4 |
| VIII | Program Elective - IV | D. Salesforce Development | 4 |

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.